

First Progress Report - Reinforcement Learning Agents for LunarLander

Work Completed

- **Environment setup and baseline interaction:** I have successfully installed and used Gymnasium to run the LunarLander environment (currently `LunarLander-v3` in code, conceptually matching the `LunarLander-v2` environment described in the earlier proposal). The `test_env.py` script verifies that the environment can be reset, stepped through with random actions, and closed cleanly, while printing observation and action space information.
- **Baseline random agent for behavior understanding:** In `random_base_agent.py`, I implemented a random-policy agent that interacts with the environment for multiple episodes, with human rendering. This gives me a qualitative understanding of the task's difficulty, typical reward ranges, and common failure modes (crashes, overshooting, etc.).

Immediate next steps

- **Align the codebase with the proposal environment and frameworks:** Standardize on `LunarLander-v3` (discrete action space) in the code, confirm Gymnasium and PyTorch are correctly installed and working together, and document exact versions plus run instructions in `README.md`.
- **Implement a first DQN baseline in PyTorch:**
 - Design a suitable neural network architecture for approximating the Q-function given the continuous state and discrete actions.
 - Implement core DQN components: replay buffer, epsilon-greedy policy, target network, and periodic target updates.
 - Integrate these into a training loop that logs episodic rewards and basic statistics.
- **Set up experiment and logging infrastructure:**
 - Add utilities to record learning curves (rewards vs. episodes, success rate estimates), track multiple random seeds, and save model checkpoints.
 - Decide on an initial set of hyper-parameters (learning rate, discount factor, batch size, replay buffer size, epsilon schedule, etc.) aligned with typical DQN settings for LunarLander.
- **Prepare the PPO implementation scaffold:**
 - Sketch the PPO software structure: policy and value networks, advantage estimation, clipping objective, and mini-batch updates.
 - Create a PPO agent skeleton file with clear interfaces (e.g., `select_action`, `update`, logging hooks) so I can implement and integrate PPO once the DQN baseline is running.

Challenges or adjustments relative to the initial proposal

- **Algorithm complexity vs. time:** Implementing both DQN and PPO “well” (with stable training, tuned hyper-parameters, and multiple seeds) is more involved than a single-algorithm project. I may need to prioritize depth over breadth-first get a strong, well-evaluated DQN baseline before adding extensive PPO variants.
- **Training stability and hyper-parameter tuning:** Both DQN and PPO can be sensitive to hyper-parameters (learning rate, clipping parameters, exploration schedules, etc.) on LunarLander. Achieving stable learning curves and reproducible performance across seeds may require more tuning runs than initially expected, which affects how many experiments and algorithmic variants fit into the timeline.
- **Discrete vs. continuous variants:** My proposal mentions experiments on both discrete and continuous action space variants. Realistically, it may be best to fully complete and analyze the discrete LunarLander-v3 first, and then treat the continuous variant as an extension if time permits, or scale down the continuous-control scope (e.g., fewer seeds or fewer algorithms).
- **Compute and runtime constraints:** Running multiple seeds for both DQN and PPO with long training horizons can be computationally expensive. I may need to adjust the number of training episodes, the number of seeds, or the number of hyper-parameter configurations to fit my available compute and time.

Overall plan to complete the project in the next month

- **Weeks of March 20-26:** Finalize PPO implementation and training. Conduct comprehensive comparison experiments between DQN and PPO across multiple random seeds for statistical robustness.
- **Weeks of March 27-April 2:** Complete literature review section. Analyze results and identify strengths and weaknesses of each approach. Begin drafting the final paper.
- **Weeks of April 3-13:** Finalize all experiments, complete the written paper including introduction, methods, results, and conclusions sections, and prepare final code submission with documentation and comments.

The project is progressing on schedule, and I am confident in completing a thorough comparison of DQN and PPO algorithms on the LunarLander environment by the April 13 deadline.